

Assignment 1.1.

Introduction to Data Cleaning

Double-click (or enter) to edit

Procedures

1. Load the dataset then show the first 5 and last 5 datapoints

```
1 from google.colab import drive
2 drive.mount('/content/drive')
3
4 import pandas as pd
5
6 df = pd.read_csv("/content/drive/MyDrive/CPE 031 /dirty_cafe_sales.csv")
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

```
1 df.head()
```

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date	
0	TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08	
1	TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023-05-16	
2	TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2023-07-19	
3	TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27	
4	TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11	

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```
1 df.tail()
```

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date	
9995	TXN_7672686	Coffee	2	2.0	4.0	NaN	UNKNOWN	2023-08-30	
9996	TXN_9659401	NaN	3	NaN	3.0	Digital Wallet	NaN	2023-06-02	
9997	TXN_5255387	Coffee	4	2.0	8.0	Digital Wallet	NaN	2023-03-02	
9998	TXN_7695629	Cookie	3	NaN	3.0	Digital Wallet	NaN	2023-12-02	
9999	TXN_6170729	Sandwich	3	4.0	12.0	Cash	In-store	2023-11-07	

QUESTION: What Column has numerical values?

The columns that has numerical values are the QUANTITY, PRICE PER UNIT, TOTAL SPENT

2. Check if there are any missing values in the dataset

```
1 df.describe()
```

	Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date	
count	10000	9667	9862	9821	9827	7421	6735	9841	
unique	10000	10	7	8	19	5	4	367	
top	TXN_9226047	Juice	5	3.0	6.0	Digital Wallet	Takeaway	UNKNOWN	
freq	1	1171	2013	2429	979	2291	3022	159	

```
1 df.isna().sum()
```

```

      0
Transaction ID    0
      Item      333
      Quantity   138
      Price Per Unit 179
      Total Spent  173
Payment Method 2579
      Location   3265
Transaction Date  159

dtype: int64

```

QUESTION: How many columns had missing values? Which column had the highest number of missing values?

7 columns have missing values, and the column LOCATION, has the highest number of missing values

3. Remove unnecessary columns and make sure numerical columns only contain numbers.

```
1 df.columns
```

```

Index(['Transaction ID', 'Item', 'Quantity', 'Price Per Unit', 'Total Spent',
      'Payment Method', 'Location', 'Transaction Date'],
      dtype='object')

```

```

1 df = df.drop(columns=['Transaction ID', 'Payment Method', 'Location', 'Transaction Date'])
2
3 df.head()

```

	Item	Quantity	Price Per Unit	Total Spent
0	Coffee	2	2.0	4.0
1	Cake	4	3.0	12.0
2	Cookie	4	1.0	ERROR
3	Salad	2	5.0	10.0
4	Coffee	2	2.0	4.0

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```
1 df.dtypes ## Checking the datatype for each column
```

```

      0
      Item    object
      Quantity object
      Price Per Unit object
      Total Spent object

dtype: object

```

```

1 df["Quantity"] = pd.to_numeric(df["Quantity"], errors='coerce')
2 df["Price Per Unit"] = pd.to_numeric(df["Price Per Unit"], errors='coerce')
3 df["Total Spent"] = pd.to_numeric(df["Total Spent"], errors='coerce')
4
5 df.head()

```

	Item	Quantity	Price Per Unit	Total Spent
0	Coffee	2.0	2.0	4.0
1	Cake	4.0	3.0	12.0
2	Cookie	4.0	1.0	NaN
3	Salad	2.0	5.0	10.0
4	Coffee	2.0	2.0	4.0



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```
1 df.dtypes ## Checking the datatype for each column
```

	0
Item	object
Quantity	float64
Price Per Unit	float64
Total Spent	float64

dtype: object

QUESTION: What happened to the object columns?

The object columns that have numerical data were converted into numeric data types, and any invalid values are replaced with NaN, meaning Not a Number, because of the use of results = 'coerce'.

4. Remove missing data to ensure all that the final dataset is clean

```
1 df.isna().sum()
```

	0
Item	333
Quantity	479
Price Per Unit	533
Total Spent	502

dtype: int64

```
1 df = df.dropna()
```

```
1 df.isna().sum()
```

	0
Item	0
Quantity	0
Price Per Unit	0
Total Spent	0

dtype: int64

```
1 df.head()
```

	Item	Quantity	Price Per Unit	Total Spent
0	Coffee	2.0	2.0	4.0
1	Cake	4.0	3.0	12.0
3	Salad	2.0	5.0	10.0
4	Coffee	2.0	2.0	4.0
5	Smoothie	5.0	4.0	20.0

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EXPLAIN the process of data cleaning done in the procedure.

** The data cleaning process began with checking the checking of the dataset for any missing values using the code line `<df.isna().sum()`, which counts the number of invalid or error entries and then computes it. We then use the line `<df.dropna()`, to remove any columns that has an error inside of it. We can then run again the `<df.isna().sum()` to recheck if all the colums are fine and do not have any errors, which we should get 0 missing values. Finally, we used the `df.head()` to view the cleaned and fixed dataset, showing the remaining rows that contains complete records without any gaps.**

Supplementary Activities

Download the dataset for supplementary activity [here](#)

1. Load the dataset and show the first 5 and last 5 datapoints
2. Check if there are any missing values in the dataset.
3. The df must only have Name, Subscribers, Category, and Country. Remove all other columns.
4. Convert Subscribers to type **int**
5. Ensure that there are no rows with missing values. (UNKNOWN is counted as a missing value)

```
1 import pandas as pd
2
3 df = pd.read_csv("/content/drive/MyDrive/CPE 031 /youtube_subscribers_data.csv")
```

```
1 df.isna().sum()
```

	0
Name	0
Brand channel	0
Subscribers (millions)	0
Primary language	0
Category	0
Country	0

dtype: int64

```
1 ## Load the dataset and show the first 5 and last 5 datapoints ##
2 df.head(5)
```

	Name	Brand channel	Subscribers (millions)	Primary language	Category	Country
0	MrBeast	No	335	English	Entertainment	United States
1	T-Series	UNKNOWN	280	Hindi	Music	India
2	Cocomelon - Nursery Rhymes	Yes	UNKNOWN	UNKNOWN	Education	United States
3	SET India	Yes	UNKNOWN	Hindi	Entertainment	India
4	Vlad and Niki	No	129	English	Entertainment	Russia

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```
1 df.tail(5)
```

	Name	Brand channel	Subscribers (millions)	Primary language	Category	Country	
45	ARY Digital HD	Yes	59.7	Urdu	Entertainment	Pakistan	
46	Fede Vigevani	No	59.4	UNKNOWN	Entertainment	Mexico	
47	LooLoo Kids	UNKNOWN	59.3	English	Music	Romania	
48	BillionSurpriseToys	Yes	UNKNOWN	English	Entertainment	United States	
49	Marshmello	No	57.5	English	Music	United States	

```
1 ## Check if there are any missing values in the dataset. ##
2 df.isna().sum()
```

	0
Name	0
Brand channel	0
Subscribers (millions)	0
Primary language	0
Category	0
Country	0

dtype: int64

```
1 df.columns
```

Index(['Name', 'Brand channel', 'Subscribers (millions)', 'Primary language', 'Category', 'Country'], dtype='object')

```
1 ## The df must only have Name, Subscribers, Category, and Country. Remove all other columns. ##
2
3 df = df.drop(columns=['Brand channel', 'Primary language'])
4 df.head()
```

	Name	Subscribers (millions)	Category	Country	
0	MrBeast	335	Entertainment	United States	
1	T-Series	280	Music	India	
2	Cocomelon - Nursery Rhymes	UNKNOWN	Education	United States	
3	SET India	UNKNOWN	Entertainment	India	
4	Vlad and Niki	129	Entertainment	Russia	

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```
1 ## Convert Subscribers to type **int** ##
2
3 df["Subscribers (millions)"] = pd.to_numeric(df["Subscribers (millions)"],
4 df.head()
```

	Name	Subscribers (millions)	Category	Country	
0	MrBeast	335	Entertainment	United States	
1	T-Series	280	Music	India	
4	Vlad and Niki	129	Entertainment	Russia	
5	Kids Diana Show	128	Entertainment	United States	Ukraine
6	Like Nastya	123	Entertainment	United States	Ukraine

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```

1  ## Ensure that there are no rows with missing values. (UNKNOWN is counted as a
   missing value) ##
2  import numpy as np
3
4  # Replace 'UNKNOWN' with NaN in 'Category' and 'Country' columns
5  df['Category'] = df['Category'].replace('UNKNOWN', np.nan)
6  df['Country'] = df['Country'].replace('UNKNOWN', np.nan)
7
8  df = df.dropna()
9
10 display(df.head())
11 display(df.isna().sum())

```

	Name	Subscribers (millions)	Category	Country	
0	MrBeast	335	Entertainment	United States	
1	T-Series	280	Music	India	
4	Vlad and Niki	129	Entertainment	Russia	
5	Kids Diana Show	128	Entertainment	United States	Ukraine
6	Like Nastya	123	Entertainment	United States	Ukraine
		0			
	Name	0			
	Subscribers (millions)	0			
	Category	0			
	Country	0			

dtype: int64

✓ Conclusion

```

1  # This activity was new to me since we were not able to create or manipulate
   data back in First year, 2nd sem. Although this activity was easy to moderate
   since I am new to the cleaning of a csv files, and that the functions are new
   to me, but doing the activity has helped me underrstand the new attributes used
   in pandas library, that focuses on the data manipulation of files.

```

Good job on finishing the activity! Kindly submit your accomplished google colab link and a PDF file of this activity.